

# Bluestock Mutual Fund Analytics Capstone Project

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Tools Used:  
Python | Pandas | NumPy | SQLite | SQL | Power BI

# Executive Summary

- Built an end-to-end mutual fund analytics platform using Python, SQLite, SQL and Power BI.

# Business Problem

- Investors need better tools to compare funds, evaluate risk and understand performance.

# Project Architecture

- Raw Data → ETL → SQLite → Analytics → Dashboard → Insights

# Dataset Overview

- Fund Master, NAV History, AUM, SIP Inflows, Transactions, Holdings and Benchmarks.

# ETL Pipeline

- Extraction, Transformation and Loading into SQLite and processed datasets.

# EDA Findings

- Industry AUM growth, SIP trends, category inflows and fund house rankings.

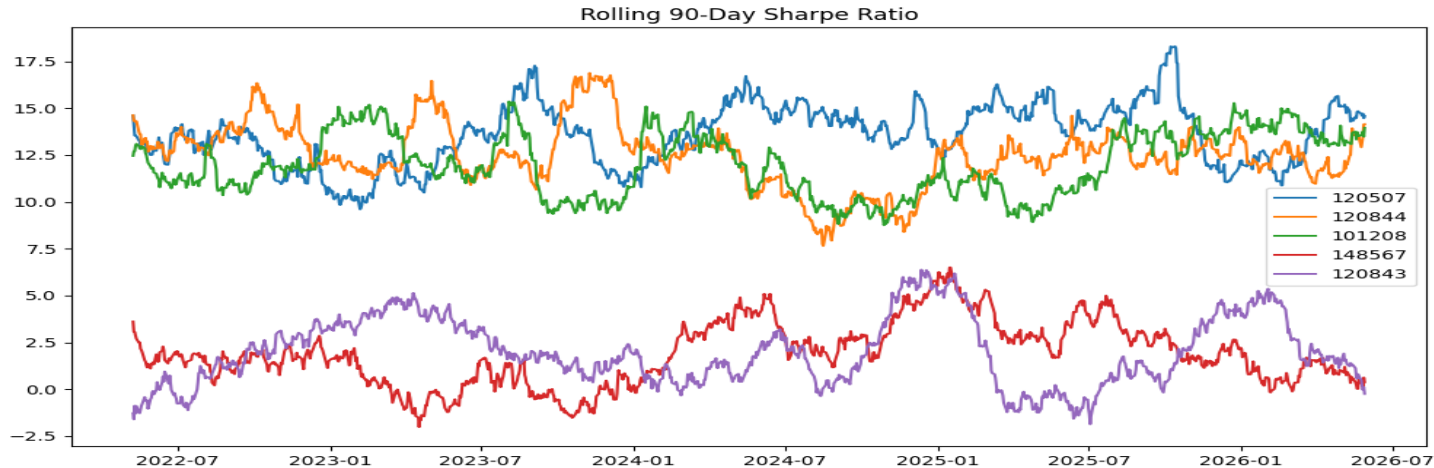
# Performance Analytics

- CAGR, Sharpe Ratio, Alpha, Beta, Maximum Drawdown and Fund Score.



# Advanced Analytics

- VaR, CVaR, Rolling Sharpe Ratio, HHI, Cohort Analysis and SIP Continuity.



- Historical VaR (95%) calculated for 40 schemes
- CVaR used to measure tail risk exposure
- Rolling 90-Day Sharpe Ratio evaluated consistency
- HHI calculated for 34 portfolios
- Investor Cohort Analysis performed
- SIP Continuity Analysis identified at-risk investors

# Investor Analytics

- Investor cohorts and SIP continuity analysis.

# Key Insights

- High Sharpe funds delivered better risk-adjusted returns; SIP participation is growing.

# Recommendations

- Focus on risk-adjusted returns, diversification and SIP retention.

# Future Scope

- Monte Carlo Simulation, Efficient Frontier and Streamlit deployment.

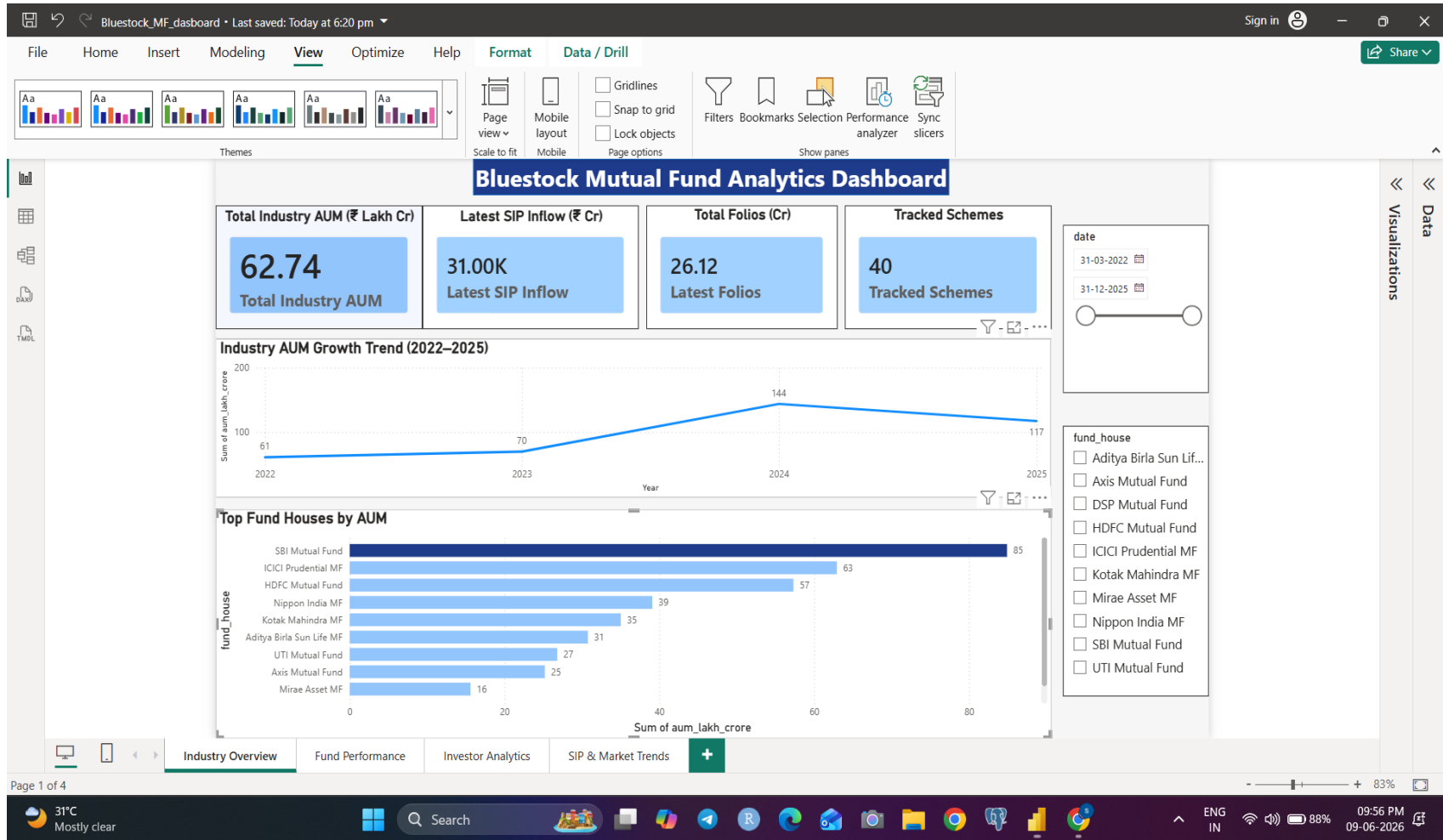
# Project Outcomes

- **40 Mutual Fund Schemes Analyzed**
- **SQLite Database Developed**
- **4 Interactive Power BI Dashboards Created**
- **VaR & CVaR Calculated**
- **Rolling Sharpe Ratio Implemented**
- **HHI Portfolio Analysis Completed**
- **Fund Recommendation Engine Developed**

# Conclusion

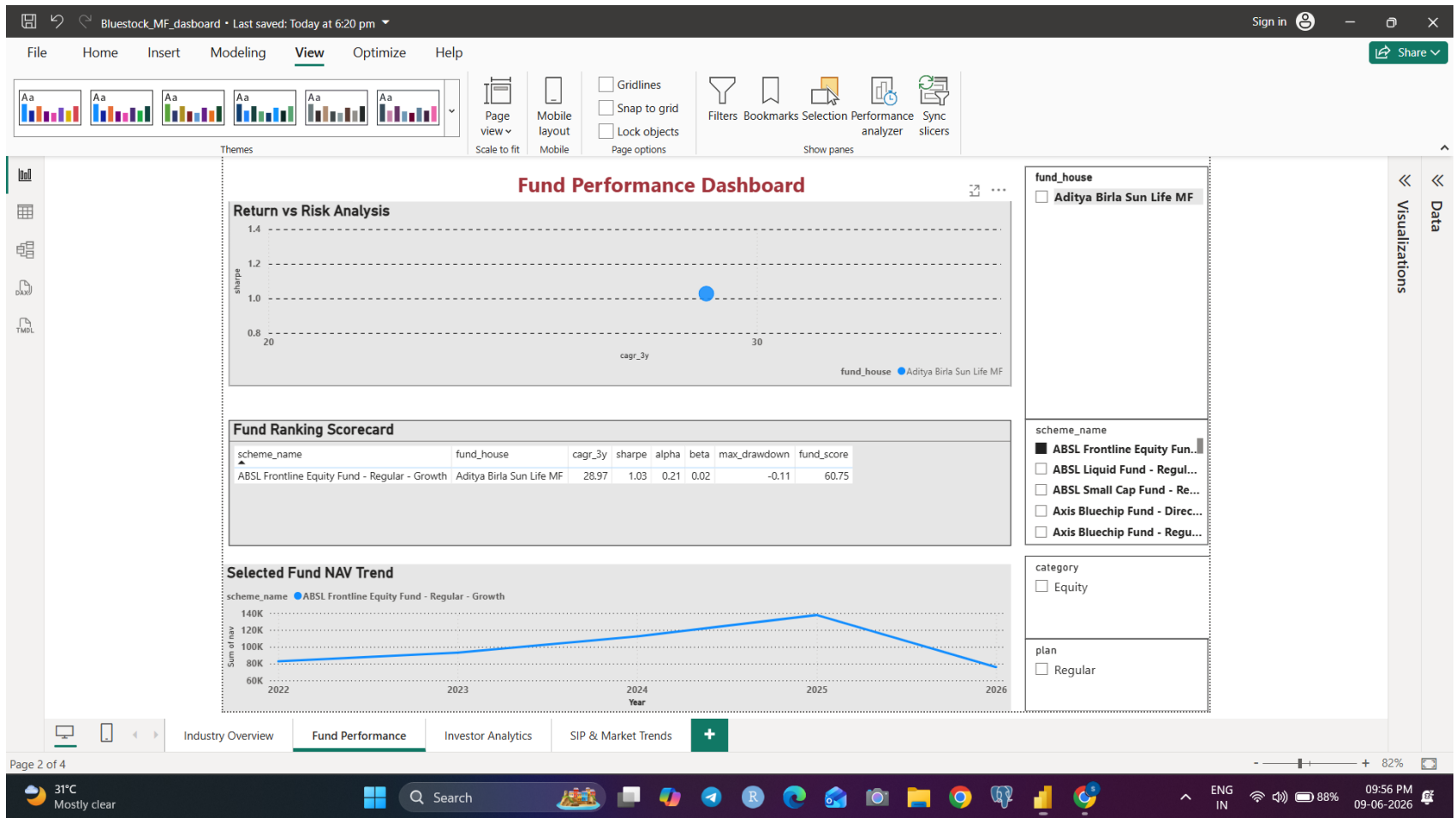
- Comprehensive mutual fund analytics solution successfully developed.

# Dashboard - Industry Overview

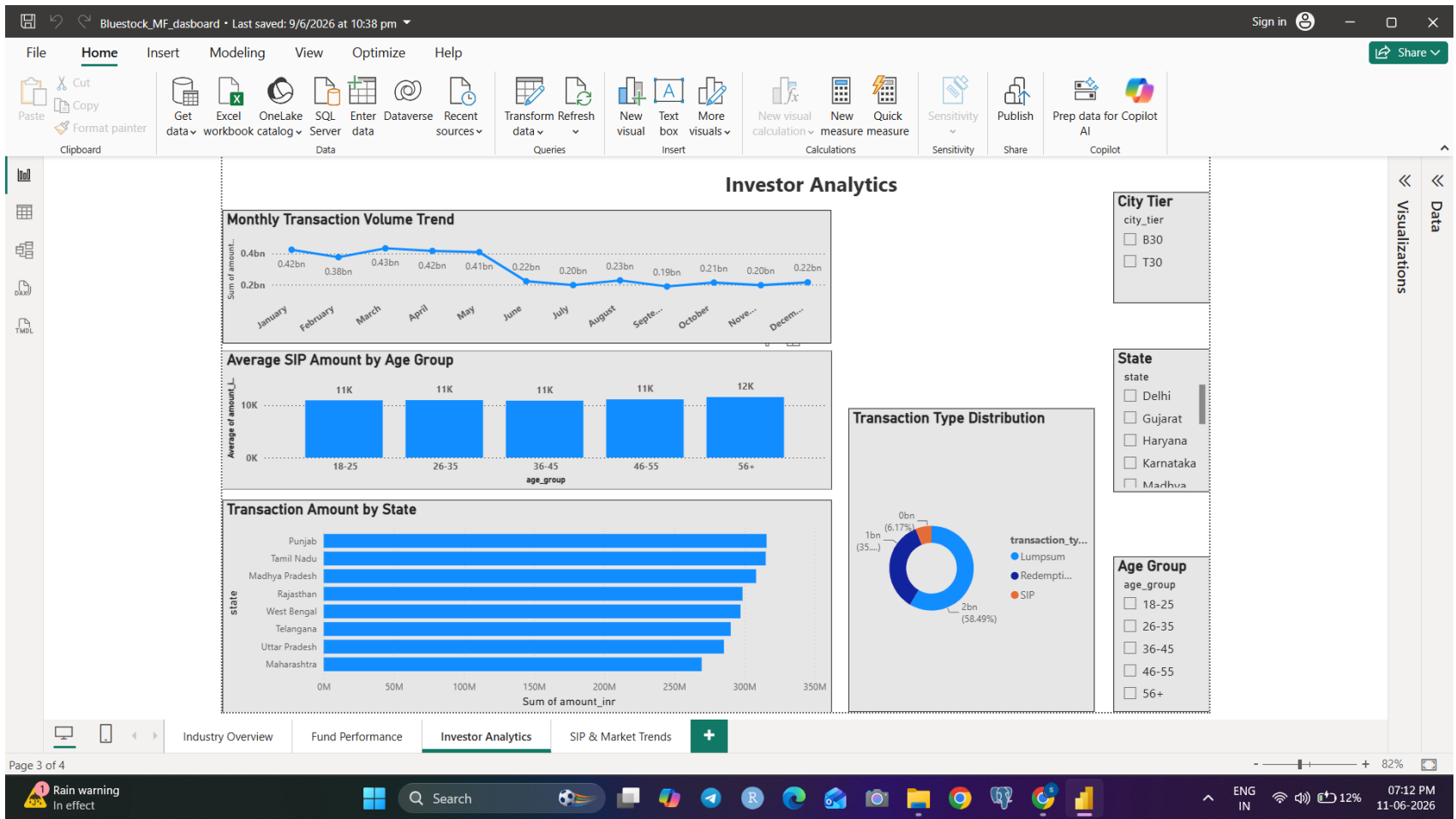




# Dashboard - Fund Performance



# Dashboard - Investor Analytics



# Dashboard - SIP & Market Trends

